

Radiosity – Global Reflection Model

Content

- ❑ Measuring illumination
- ❑ Radiation equation
- ❑ Form factors computation
- ❑ Radiosity

Local and global illumination

- ❑ Local illumination: Phong model (used in OpenGL)
 - ❑ Light ray reflected by surface to viewer
 - ❑ No shadows on the neighbor objects, no multiple reflections
 - ❑ Good quality and fast enough for interactive graphics
- ❑ Global illumination: Ray tracing
 - ❑ Multiple specular reflections and transmissions
 - ❑ Simple diffuse reflection (just one step)
 - ❑ Shine and transparent objects
- ❑ Global illumination: Radiosity
 - ❑ All multiple diffuse reflections
 - ❑ Cannot represent transparent objects
 - ❑ May add specular reflection -> advanced method

Computation space

Ray tracing and Radiosity make computations in different coordinate spaces.

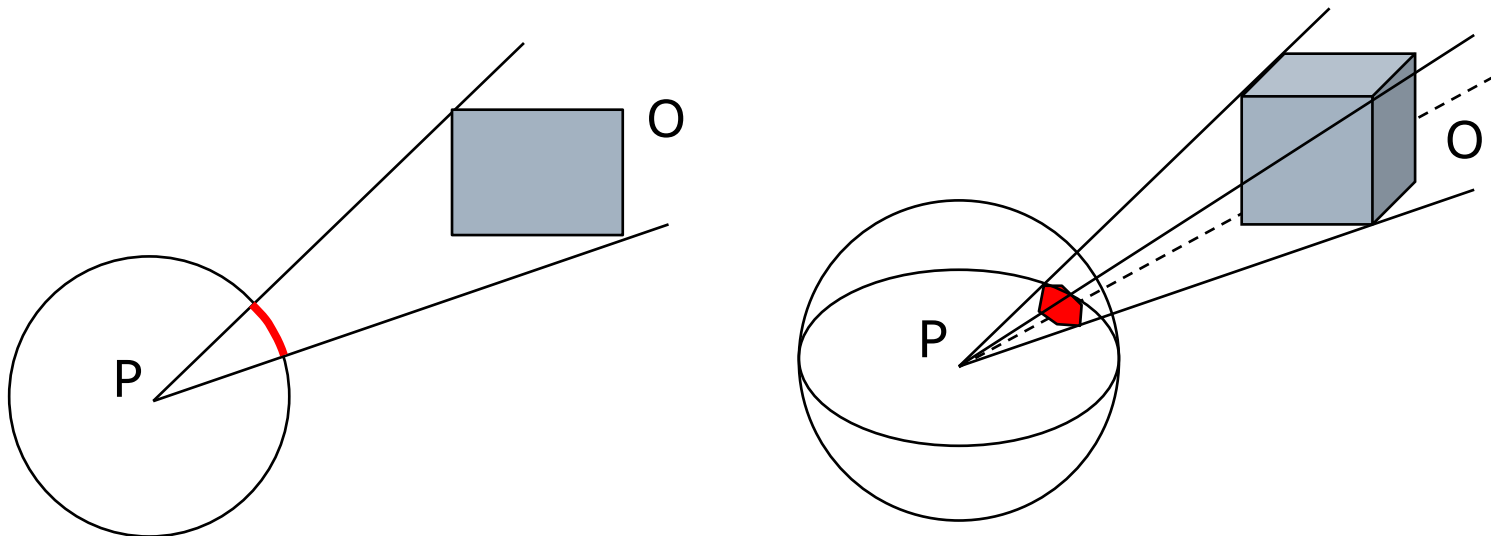
- ❑ Image space: Ray tracing
 - ❑ Trace the view ray, from viewer
 - ❑ Computaton dependent on viewer position
 - ❑ Finally get a raster image, compute each pixel of the matrix
- ❑ Object space: Radiosity
 - ❑ Consider just the diffuse interaction among objects
 - ❑ Computation independent on viewer position
 - ❑ Finally get an improved 3D model, the color for each surface patch
 - ❑ OpenGL as render for the improved 3D model

Radiosity principle

- ❑ Split the object surface into patches (i.e. elements)
- ❑ Describe the transfer of light between patches by a system of linear equations
- ❑ Assumptions:
 - ❑ Reflection and emission are diffuse
 - Diffuse reflection is uniform toward all directions
 - Radiance is independent of direction
 - ❑ Does not model the space transparency (e.g. No fog)
 - ❑ No transparent objects -> no transmitted rays
 - ❑ Radiosity is constant across each patch (i.e. element)
 - ❑ For color make computations for each component R, G, and B

2D and 3D solid angles

- ❑ 2D angle is subtended by object O from point P:
 - ❑ Given by the length of the arc on the unit circle of P
 - ❑ Measured in radians ($0 \dots 2\pi$)
- ❑ 3D solid angle is subtended by object O from point P:
 - ❑ Given by the spherical area on the unit sphere of P
 - ❑ Measured in radians ($0 \dots 4\pi$)



Radiant power and Radiosity

- *Radiant power* **P**:

Rate of light energy transmission

power = energy / time

- *Flux density* ϕ :

Radiant power per unit surface

flux density = power / area

- *Irradiance* **E**: Incident flux density of surface

- *Radiosity* **B**: exitant flux density of surface

radiosity = power / area

- *Flux density at a point:*

$$\phi(x) = dP/d(x)$$

Power at point toward a direction

- *Radiant intensity* **I**:

Power radiated per unit solid angle by point surface

radiant intensity = power / solid angle

- *Radiant intensity in direction* ω :

$$I(\omega) = dP/d\omega$$

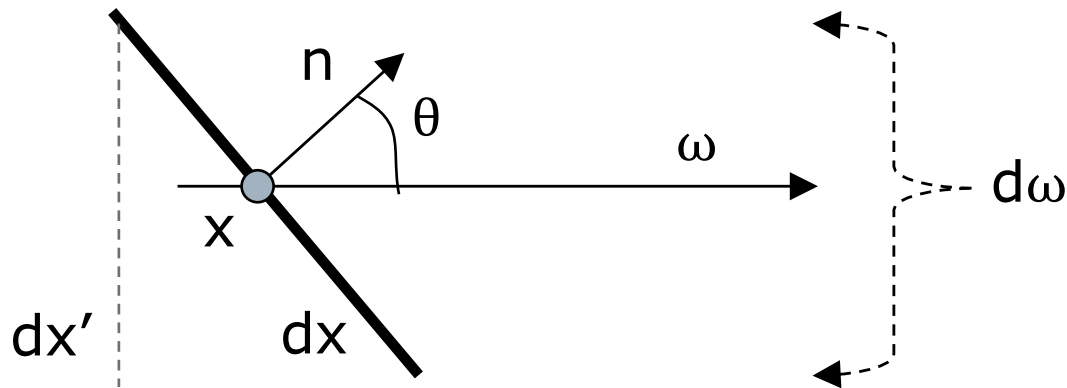
- *Radiance* **L(x, ω)**

Flux density at point x in direction ω

radiance = power / (area * solid angle)

Radiance

- Measured across surface at point x in direction ω :



$$L(x, \omega) = \frac{d^2 P}{d\omega dx'} = \frac{d^2 P}{d\omega \cos \theta dx}$$

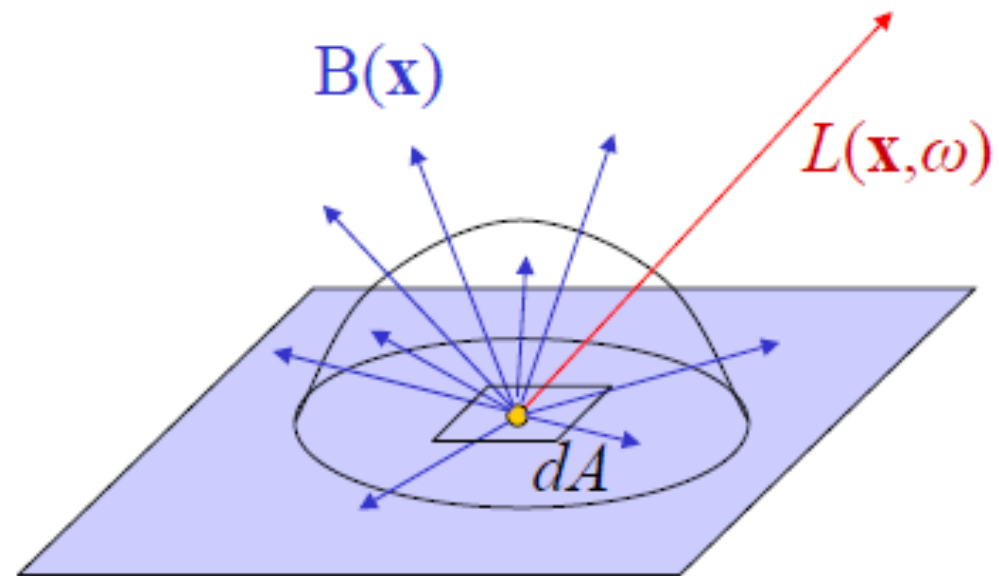
Radiosity and Radiance

Radiosity: $B(x) = dP / dx$

Radiance: $L(x, \omega) = d^2P / d\omega \cos\theta dx$

Let Ω be the set of all directions above x :

$$B(x) = \int_{\Omega} L(x, \omega) \cos\theta d\omega$$

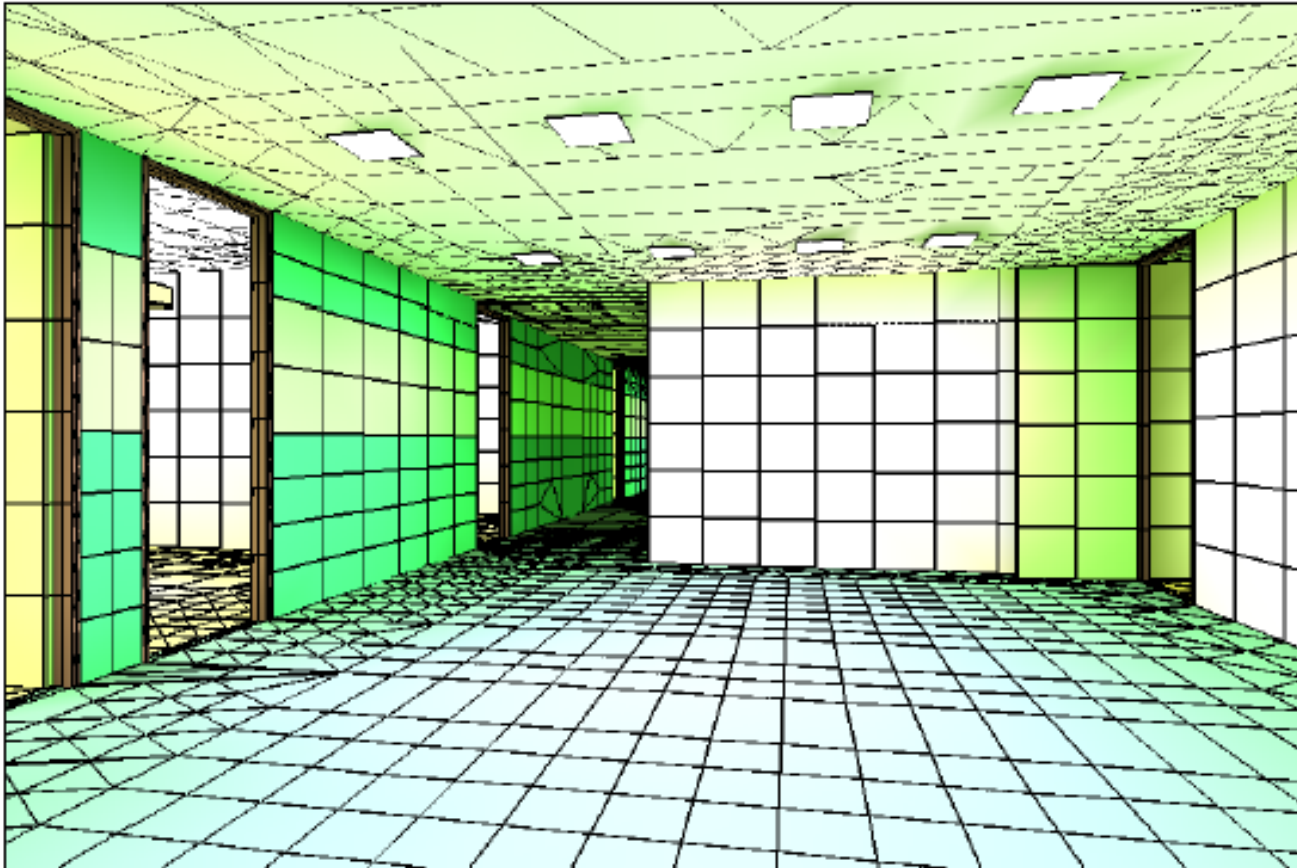


Radiosity equation– Energy balance

- Lambertian surface, ideal diffuse surface
- Split on n elements
- Variables:
 - A_i** Area of element i (can be computed)
 - B_i** Radiation of element i (unknown)
 - E_i** Radiant emitted flux density of element i (given)
 - ρ_i** Reflectance of element i (given)
 - F_{ji}** Form factor from j to i (can be computed)

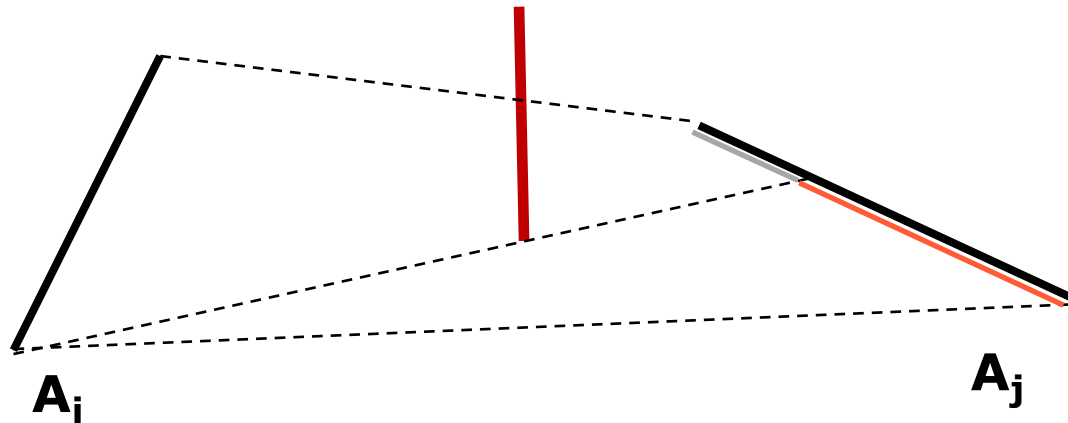
$$A_i B_i = A_i E_i + \rho_i \sum_{j=1}^n F_{ji} A_j B_j$$

Mesh surfaces into elements

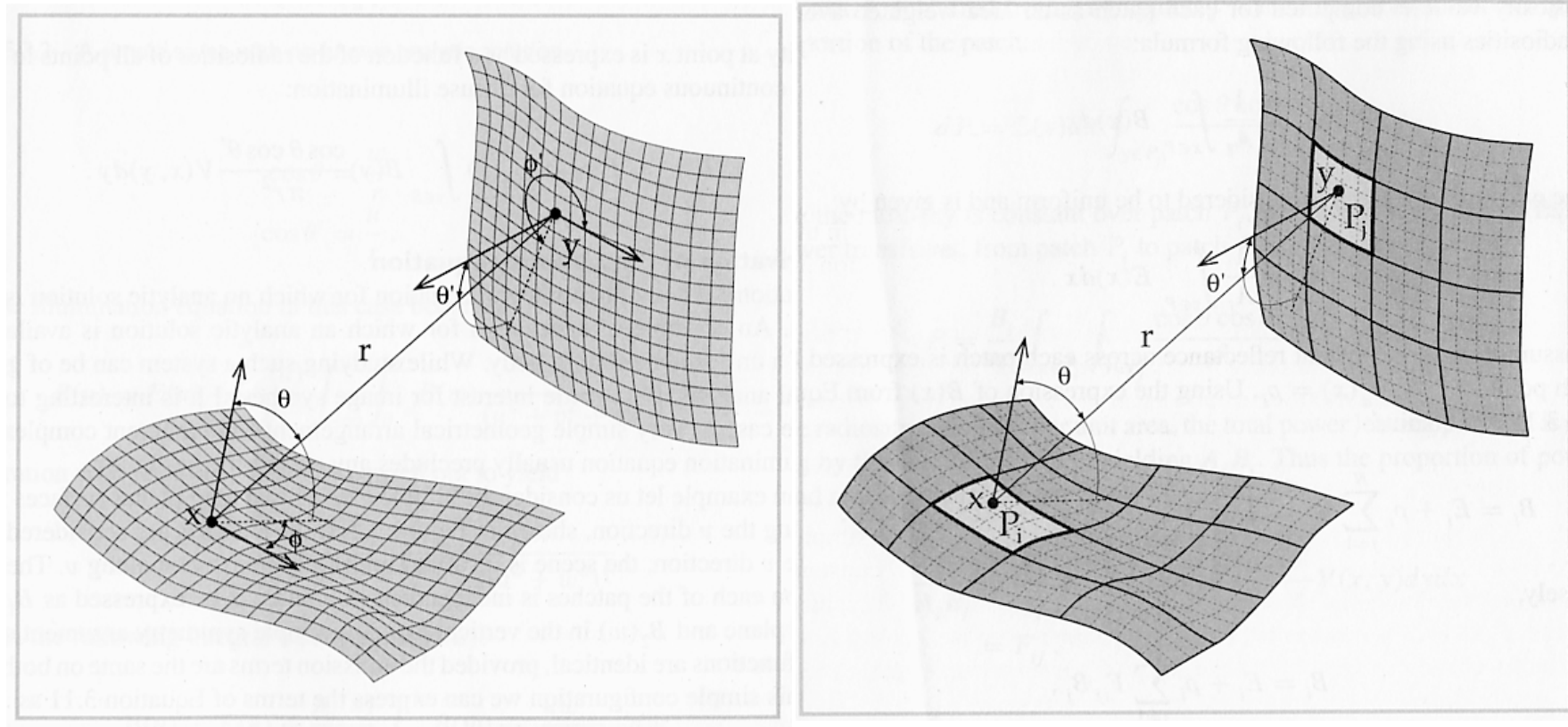


Form factors

- Form factor F_{ij} is the fraction of light leaving element i and arriving at element j .
- Form factor depends on:
 - Shape of elements i and j
 - Relative orientation of elements i and j
 - Distance between the two elements
 - Occlusion by other elements



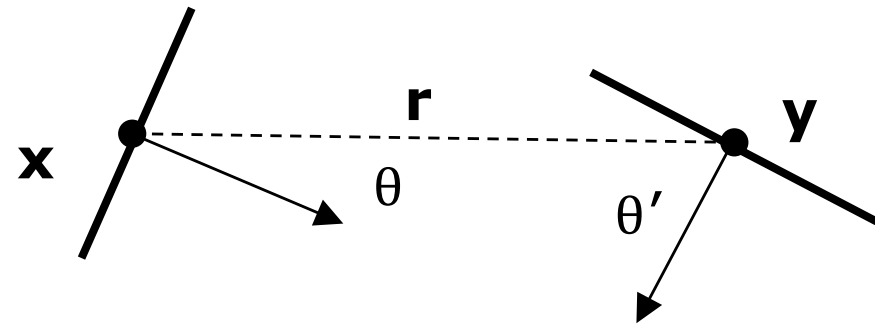
Form factors for pair of patches



Form factor equation

□ Let us define:

- Polar angles θ and θ' between elements' normals and light ray between elements x and y .
- Visibility function $v(x, y) = 0$, if the light ray between x and y is occluded, otherwise $v(x, y) = 1$.
- Distance between x and y is r .



$$A_i F_{ij} = \int_{x \in P_i} \int_{y \in P_j} \frac{\cos \theta \cos \theta'}{\pi r^2} v(x, y) dy dx$$

Symetry of form factor

Symetric definition:

$$A_i F_{ij} = \int_{x \in P_i} \int_{y \in P_j} \frac{\cos \theta \cos \theta'}{\pi r^2} v(x, y) dy dx = A_j F_{ji}$$

Radiosity equation:

$$A_i B_i = A_i E_i + \rho_i \sum_{j=1}^n F_{ji} A_j B_j$$

Devided by A_i to get the expresion of radiation B_i :

$$\begin{aligned} B_i &= E_i + \rho_i \sum_j (F_{ji} A_j / A_i) B_j \\ &= E_i + \rho_i \sum_j F_{ij} B_j \end{aligned}$$

System of linear equations

Rewrite the radiosity equation: $B_i - \rho_i \sum_j F_{ij} B_j = E_i$

As a matrix:

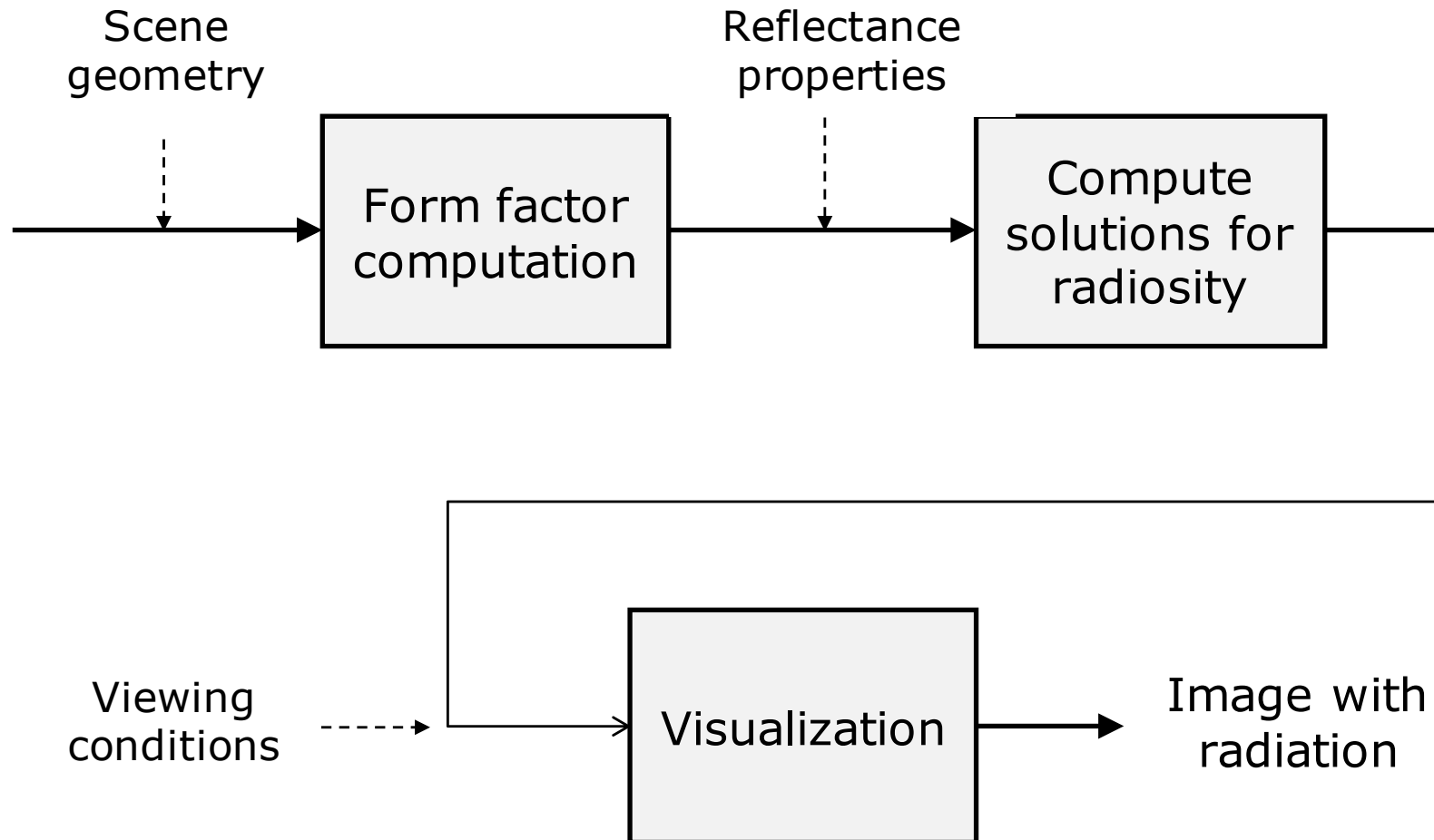
$$\begin{bmatrix} 1 - \rho_1 F_{11} & -\rho_1 F_{12} & \cdots & \rho_1 F_{1n} \\ -\rho_2 F_{21} & 1 - \rho_2 F_{22} & \cdots & \rho_2 F_{2n} \\ \vdots & \vdots & & \vdots \\ -\rho_n F_{n1} & \rho_n F_{n2} & \cdots & 1 - \rho_n F_{nn} \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \\ \vdots \\ B_n \end{bmatrix} = \begin{bmatrix} E_1 \\ E_2 \\ \vdots \\ E_n \end{bmatrix}$$

Given: reflectances ρ_i , form factors F_{ij} , emissions E_i

Unknown: radiosities B_i

=> n equations with n unknown

Computing radiosity



Scene visualization

- ❑ Radiosity solution does not depend on visualization
- ❑ Hardware modules may support performant computation
- ❑ Color of the patches will be associated to vertices
- ❑ The processing can be parallelized

Form factor computation

- Analyze of intervisibility between elements is a critical phase:
 - Many cases can be resolved analitically
 - Avoid computation of double integral
 - There are two main computation approaches:
 1. Hemicube: by z-buffer method. Can be supported by hardware
 2. Ray casting (slower)
- Both approaches show aliasing effects.

Radiosity equations

Radiosity equation:
$$B_i - \rho_i \sum_j F_{ij} B_j = E_i$$

Matrix equation:

$$\begin{bmatrix} 1 - \rho_1 F_{11} & -\rho_1 F_{12} & \cdots & \rho_1 F_{1n} \\ -\rho_2 F_{21} & 1 - \rho_2 F_{22} & \cdots & \rho_2 F_{2n} \\ \vdots & \vdots & & \vdots \\ -\rho_n F_{n1} & \rho_n F_{n2} & \cdots & 1 - \rho_n F_{nn} \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \\ \vdots \\ B_n \end{bmatrix} = \begin{bmatrix} E_1 \\ E_2 \\ \vdots \\ E_n \end{bmatrix}$$

Unknown: radiosities B_i

Given: reflectances ρ_i , form factors F_{ij} , emissions E_i

Classical radiosity algorithms

1. Matrix radiosity:

- Diagonally dominant matrix
- Use Gauss-Seidel iterative solution
- Time and space complexity is $O(n^2)$, for n elements
- Use extensive memory

2. Progressive refinement method:

- Solve equations incrementally
- Time complexity is $O(n-s)$ for s iterations
- Space complexity is $O(n)$

Progressive refinement

1. Compute all form factors F_{ij}
2. Consider initial approximation of the radiosity:
 - Emitting elements have $B_i = E_i$
 - Other elements have $B_i = 0$
3. Compute the next approximation for radiosity:

$$B'_i = E_i + \rho_i \sum_j F_{ij} B_j$$

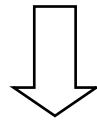
4. Iterate with the new approximation

Conclusions

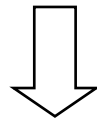
- ❑ Assumptions:
 - ❑ Opaque Lambertian surfaces
 - ❑ Constant radiation across each element
- ❑ Radiation computation:
 1. Spit the scene surfaces into patches
 2. Compute form factors
 3. Compute radiation by solving the linear equations
 4. Render the 3D model by standard hardware

Classical radiosity algorithm

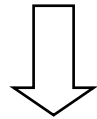
Mesh surfaces into elements



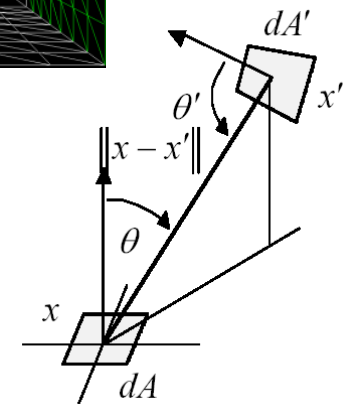
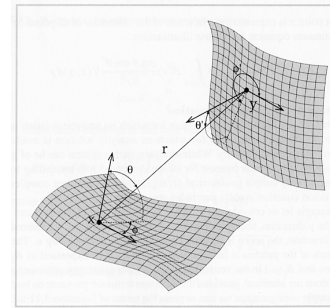
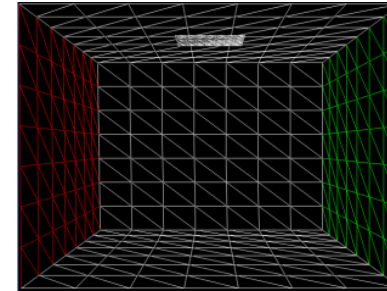
Compute form factors between elements



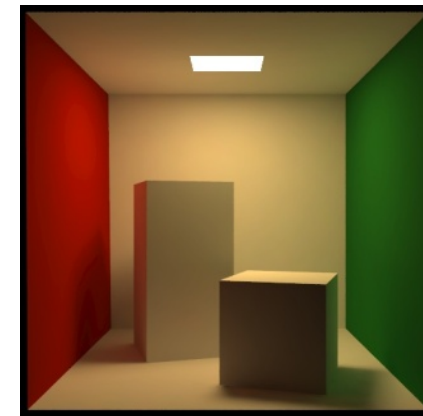
Solve linear system for radiosities



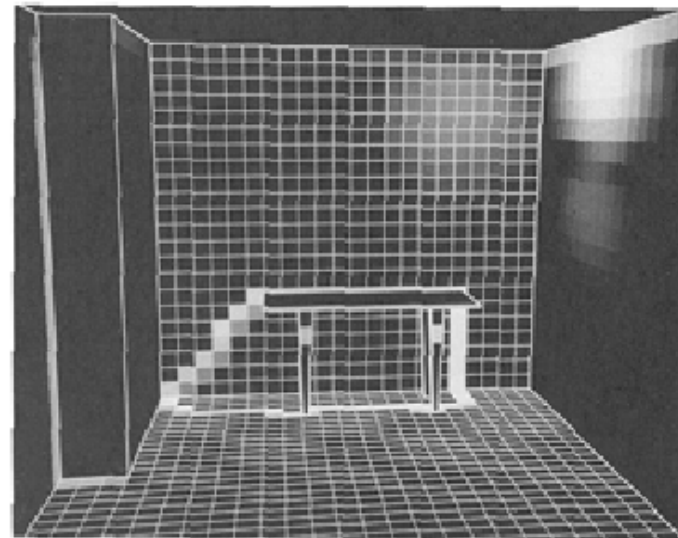
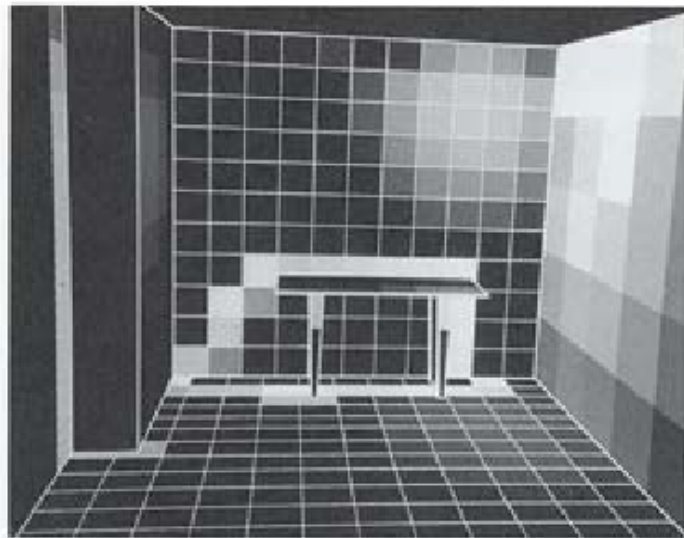
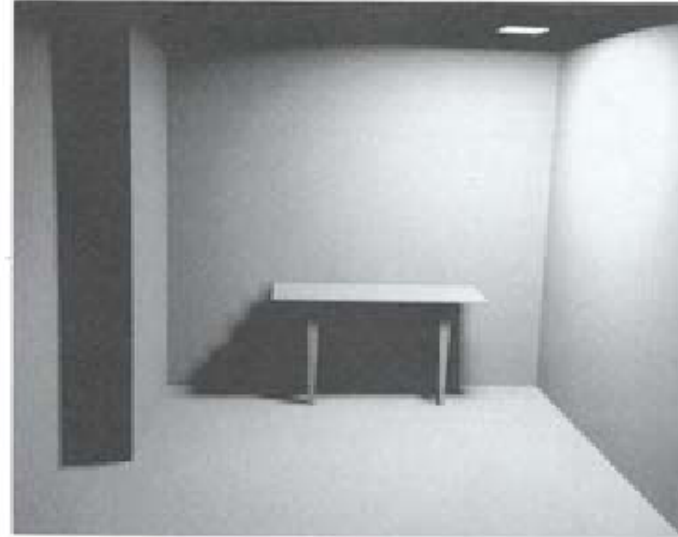
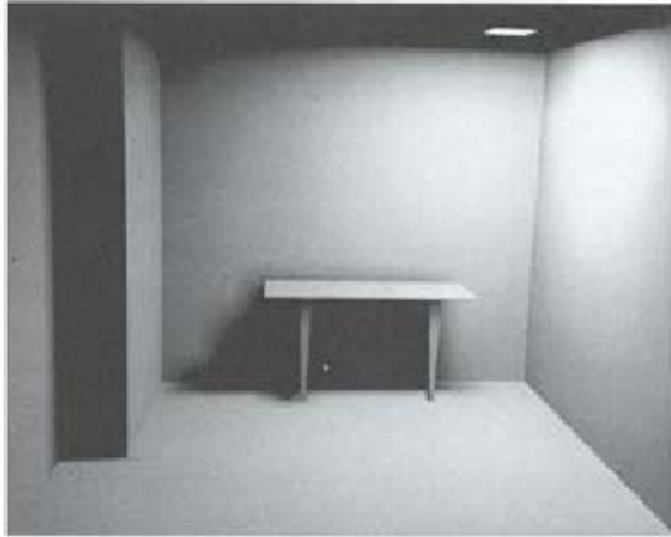
Reconstruct and display solution



$$\begin{bmatrix} R_1 \\ R_2 \\ R_3 \\ R_4 \\ R_5 \\ R_6 \\ R_7 \\ R_8 \\ R_9 \\ R_{10} \end{bmatrix} = \begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{14} \\ F_{15} \\ F_{16} \\ F_{17} \\ F_{18} \\ F_{19} \\ F_{10} \end{bmatrix} + \begin{bmatrix} E_1 \\ E_2 \\ E_3 \\ E_4 \\ E_5 \\ E_6 \\ E_7 \\ E_8 \\ E_9 \\ E_{10} \end{bmatrix}$$



Radiosity – increasing resolution



Ray tracing examples



Radiosity examples

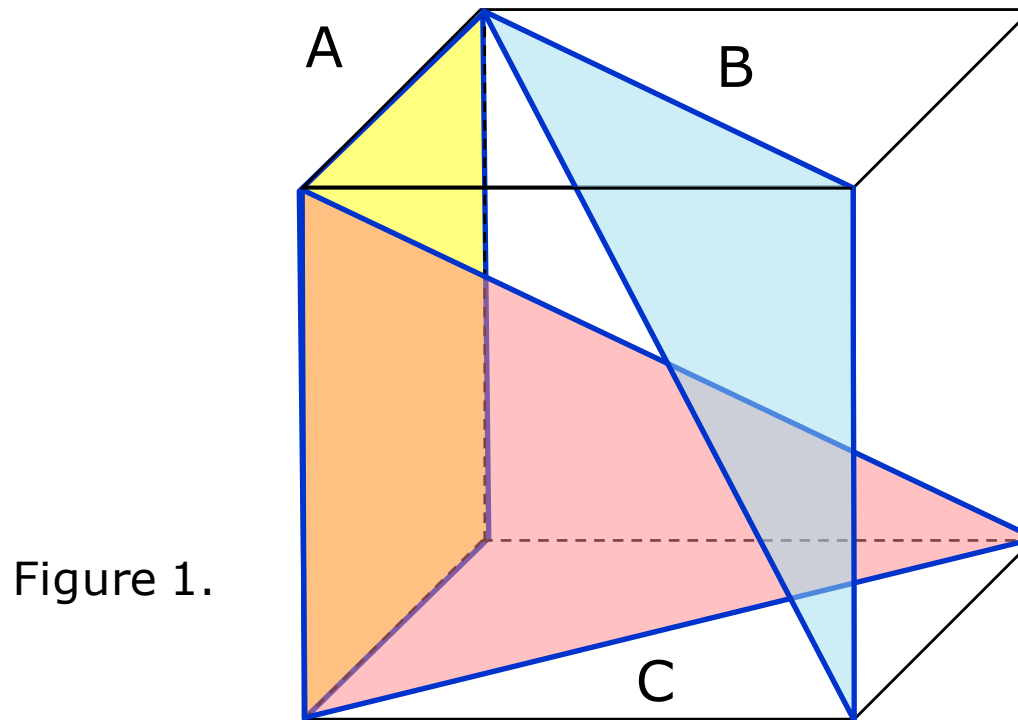


Ray tracing vs Radiosity examples



Questions and problems

1. Exemplify and explain the Radiosity method for the object in figure 1. The scene of objects consists of: a) entire vertical rectangle A is a light source; b) triangle B blue, and C red are opaque.



Questions and problems

2. Analyze and compare the two scenes rendered by Ray tracing and by Radiosity methods. Highlight and explain the differences.
3. Explain the Progressive refinement method on the scene in figure 1.